Installing the Jetson component environment

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Some users may need to use the system component environment that comes with NVIDIA. After the system has been installed and you have successfully entered the system desktop, you can install the component environment by following the tutorial below!

Note: The SSD bundled with the product already contains the factory image, and the factory image includes the official NVIDIA system and the mainboard environment; do not proceed unless you specifically need the clean system!

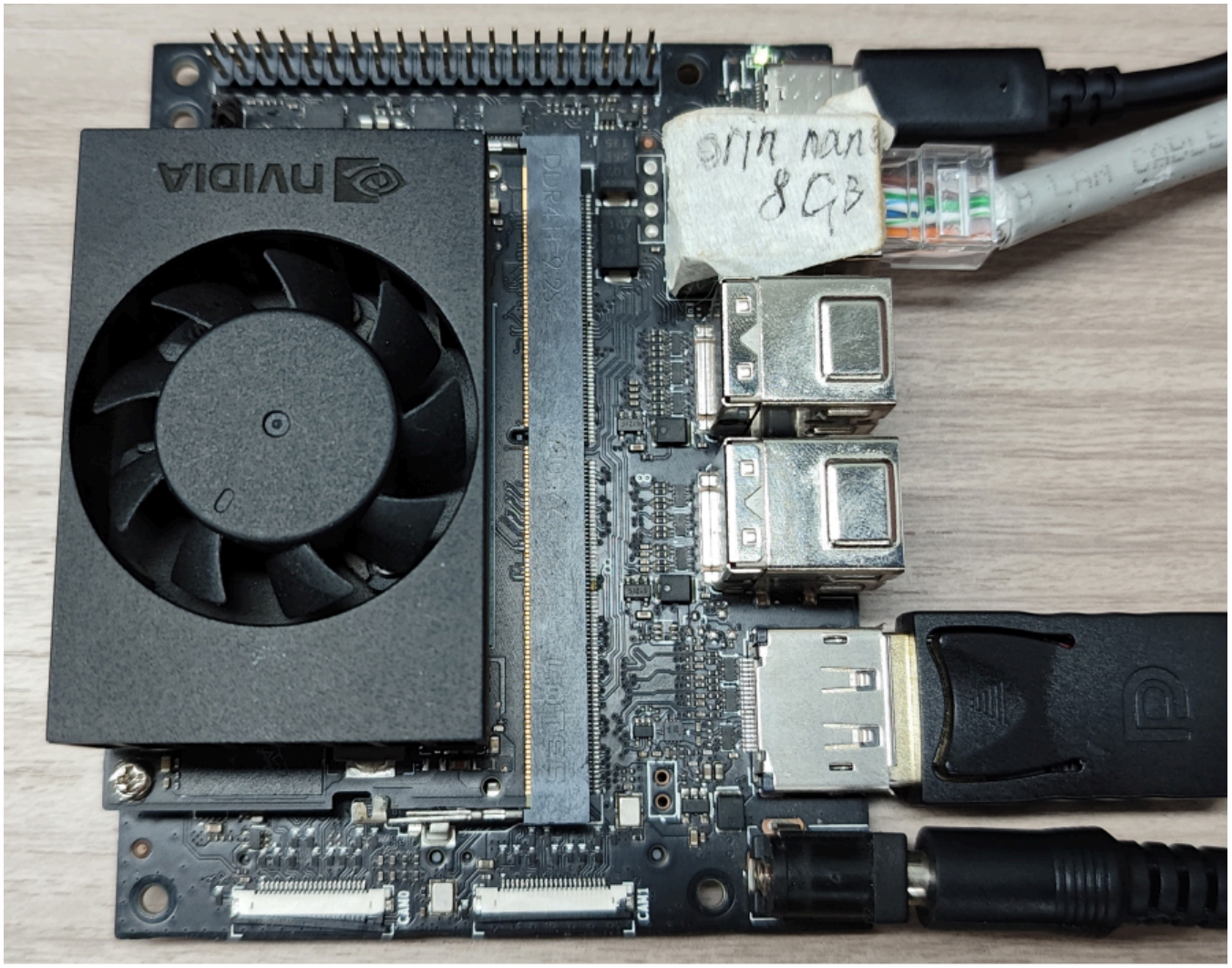
```
The whole process requires a network connection, and some components may require a proxy; if you cannot complete the component installation, use our factory image
```

1. Hardware connections

The Jetson Orin mainboard must be connected to a DC power adapter, a DP cable, an Ethernet cable and a Type C cable: connect the Type C cable to the computer or the virtual machine

The illustration below uses the official Jetson Orin version as an example; users of other versions can refer to it (the hardware interfaces and functional layout are identical)

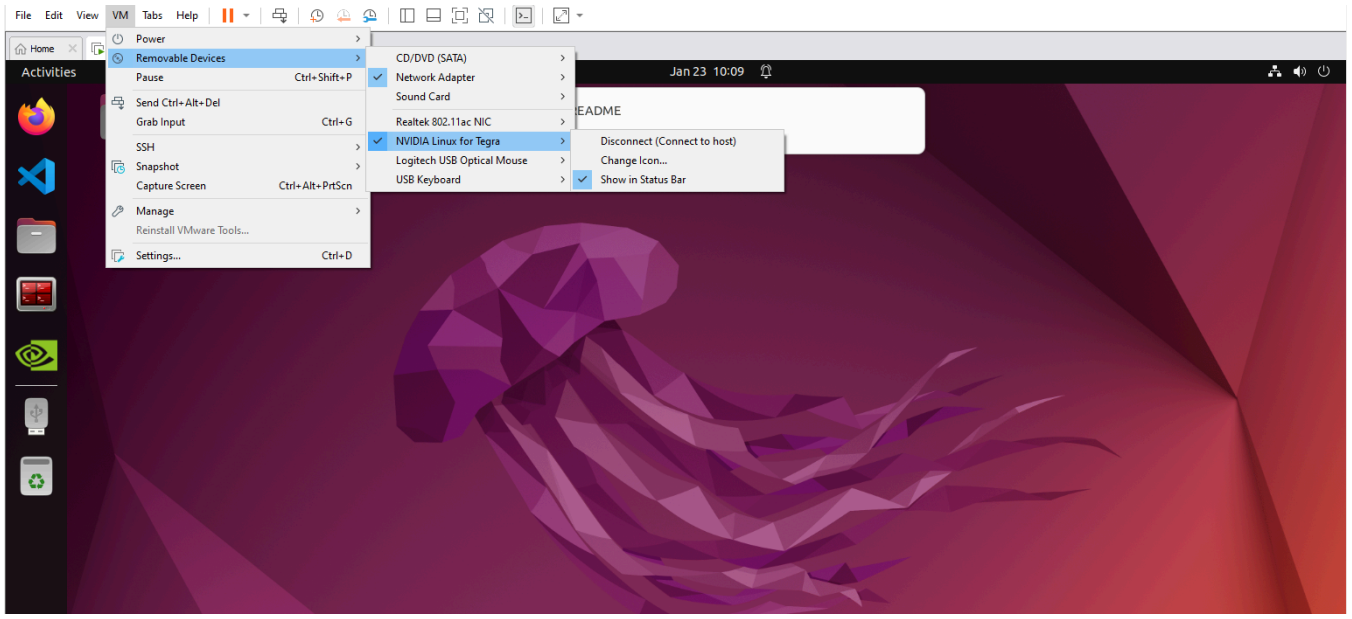
Note: Installing the Jetson component environment does not require entering flash mode; the mainboard only needs to be powered on!!!



2. Software connection

This tutorial uses VMware Workstation 17 Pro to flash the system!

After starting the virtual machine, select in the menu bar: Virtual Machine → NVIDIA Linux for Tegra → confirm the connection status with the virtual machine



3. Using SDK Manager

The Jetson Orin mainboard is in the normal system boot state; connect it to the computer or the virtual machine with a Type-C cable.

3.1. Mainboard selection

After opening `SDK Manager`, select according to your mainboard model: for the official kit choose `Jetson Orin NX 16GB`

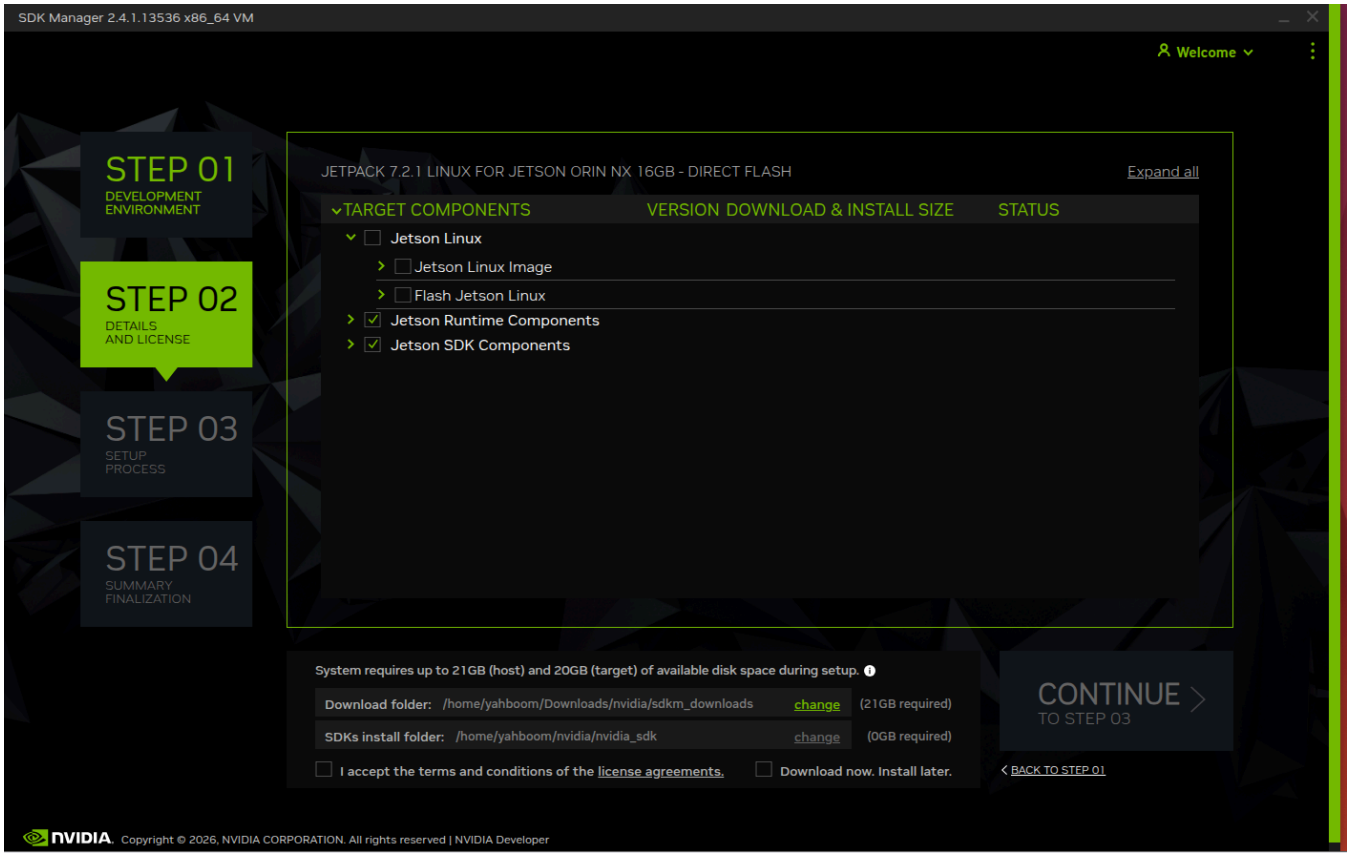
3.2. STEP1

Tick the options as shown below; after confirming that the selections match, click `CONTINUE`:

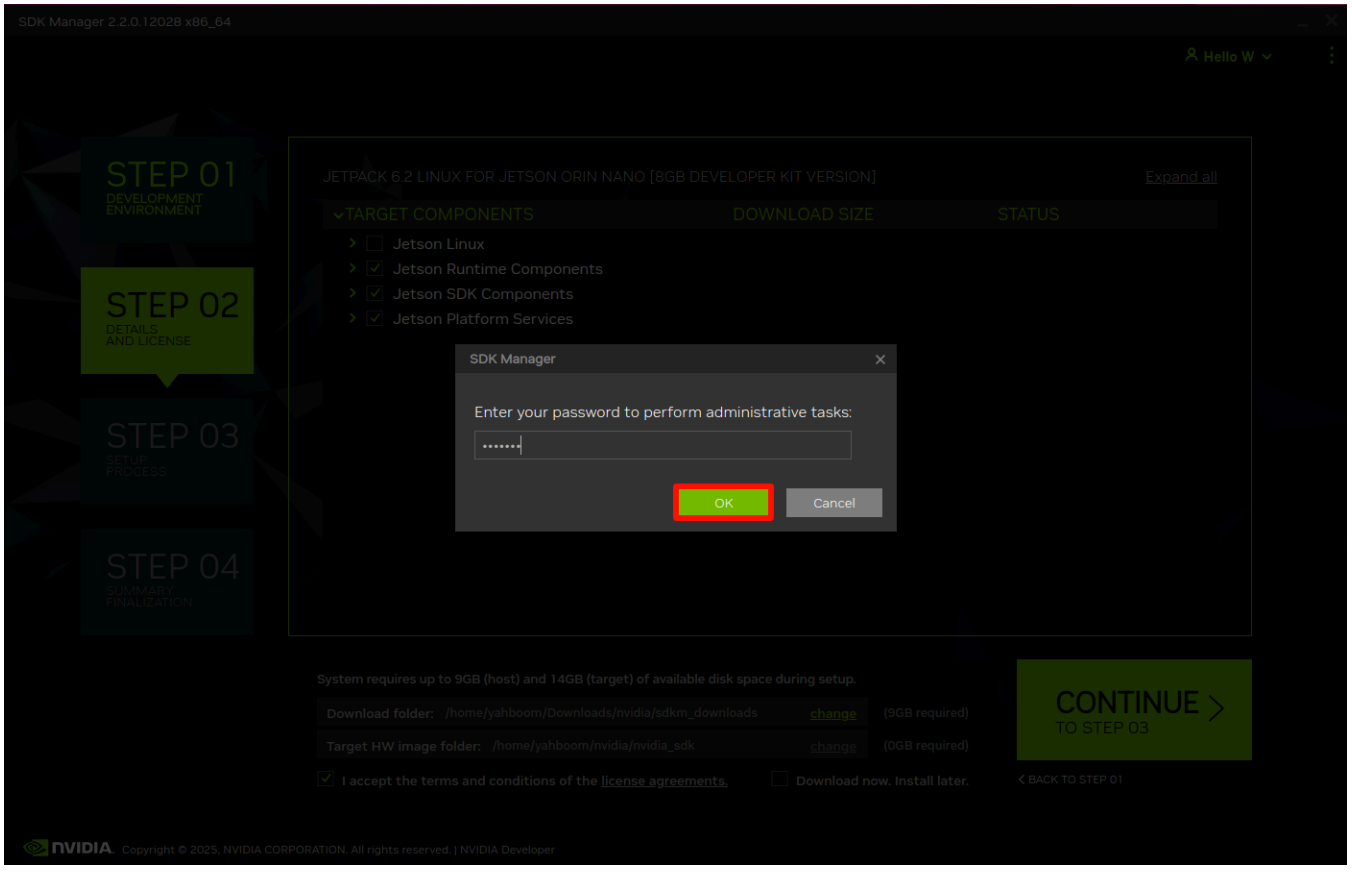


3.3. STEP2

The system has already been installed successfully; we only need to tick the component environments we need. SDK Manager will download all the components automatically, and once the download is complete you can continue to the next step

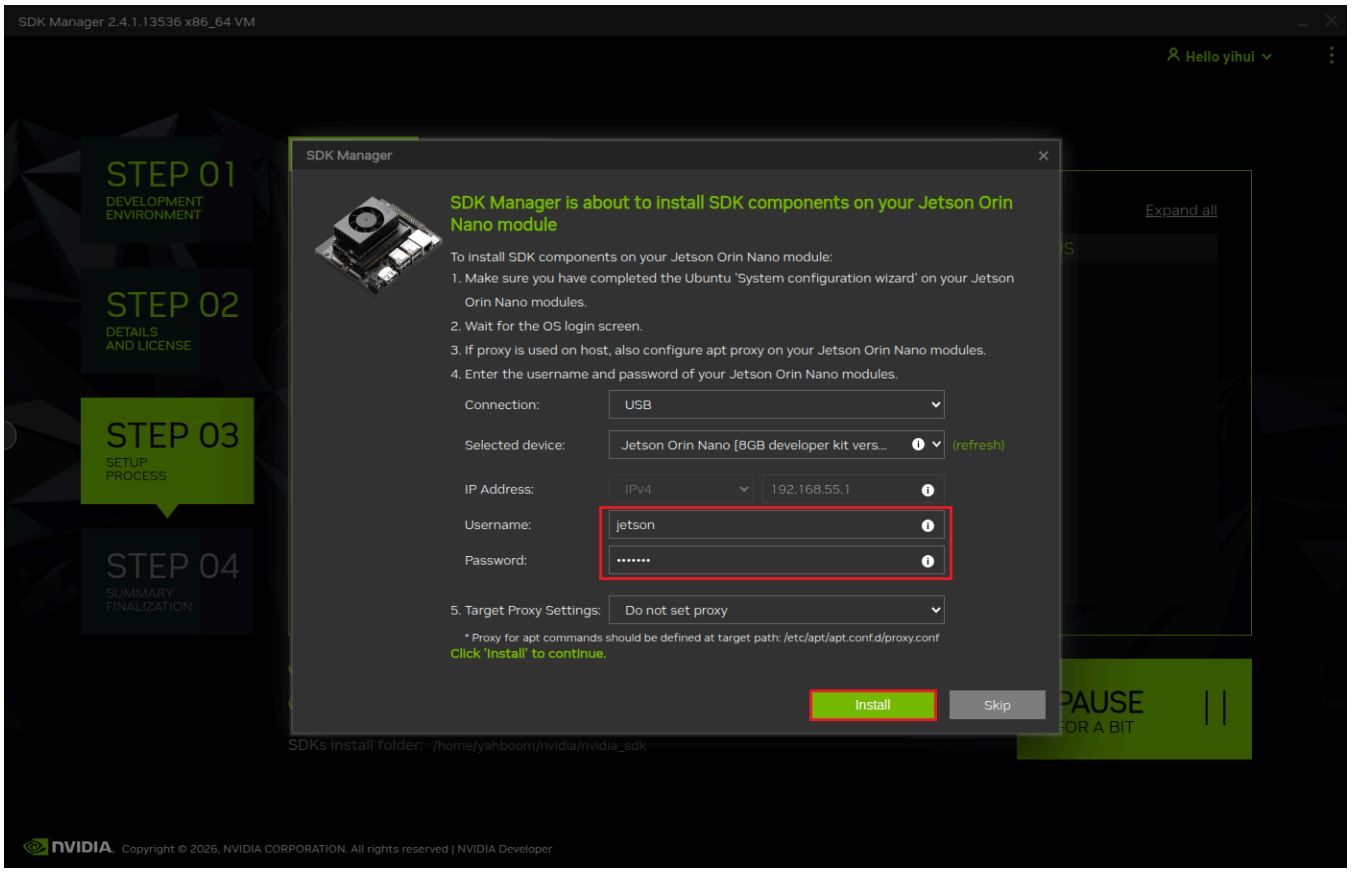


Enter the virtual machine password: yahboom; if it is your own virtual machine, enter your own password

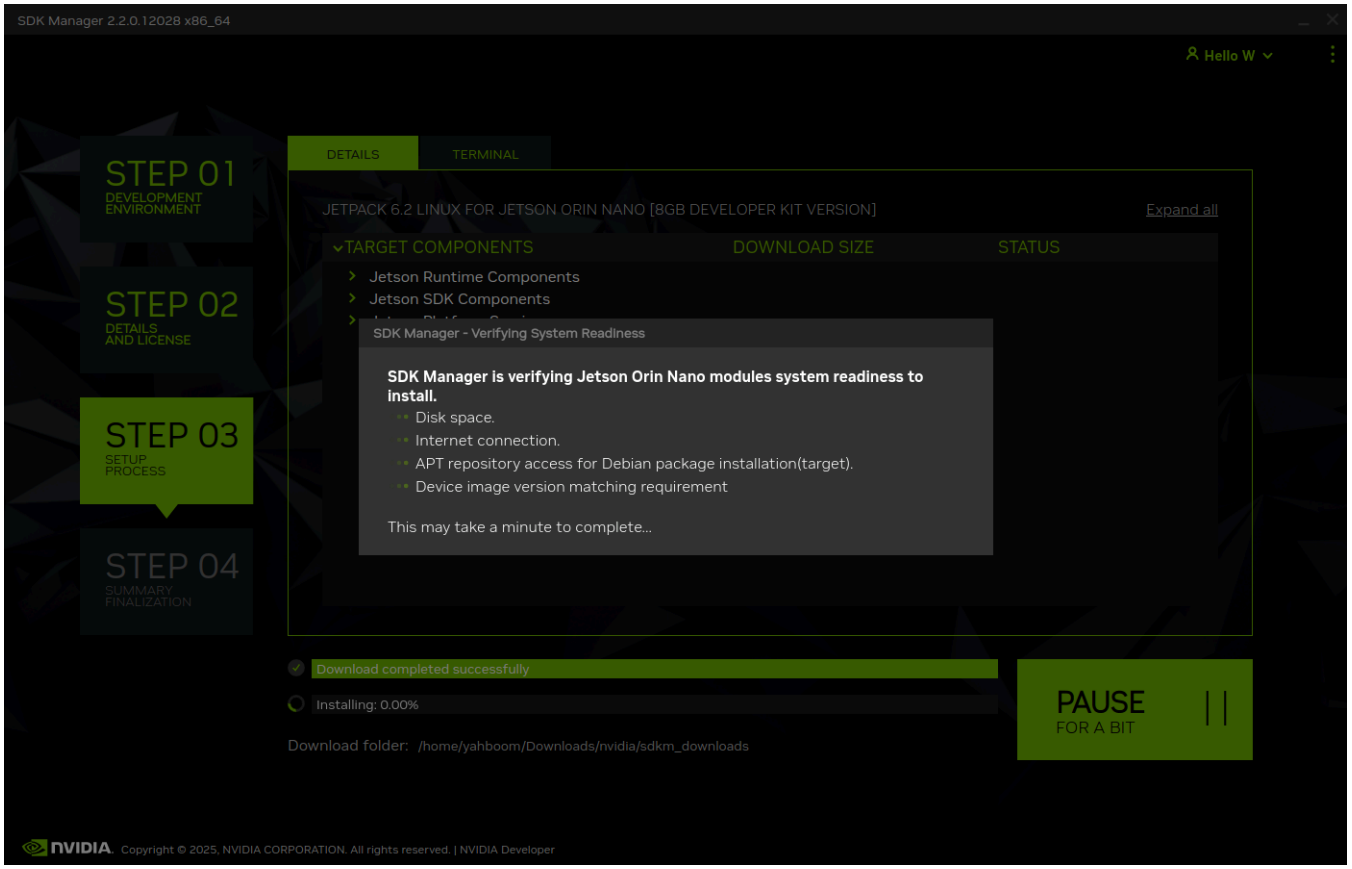


3.4. STEP3

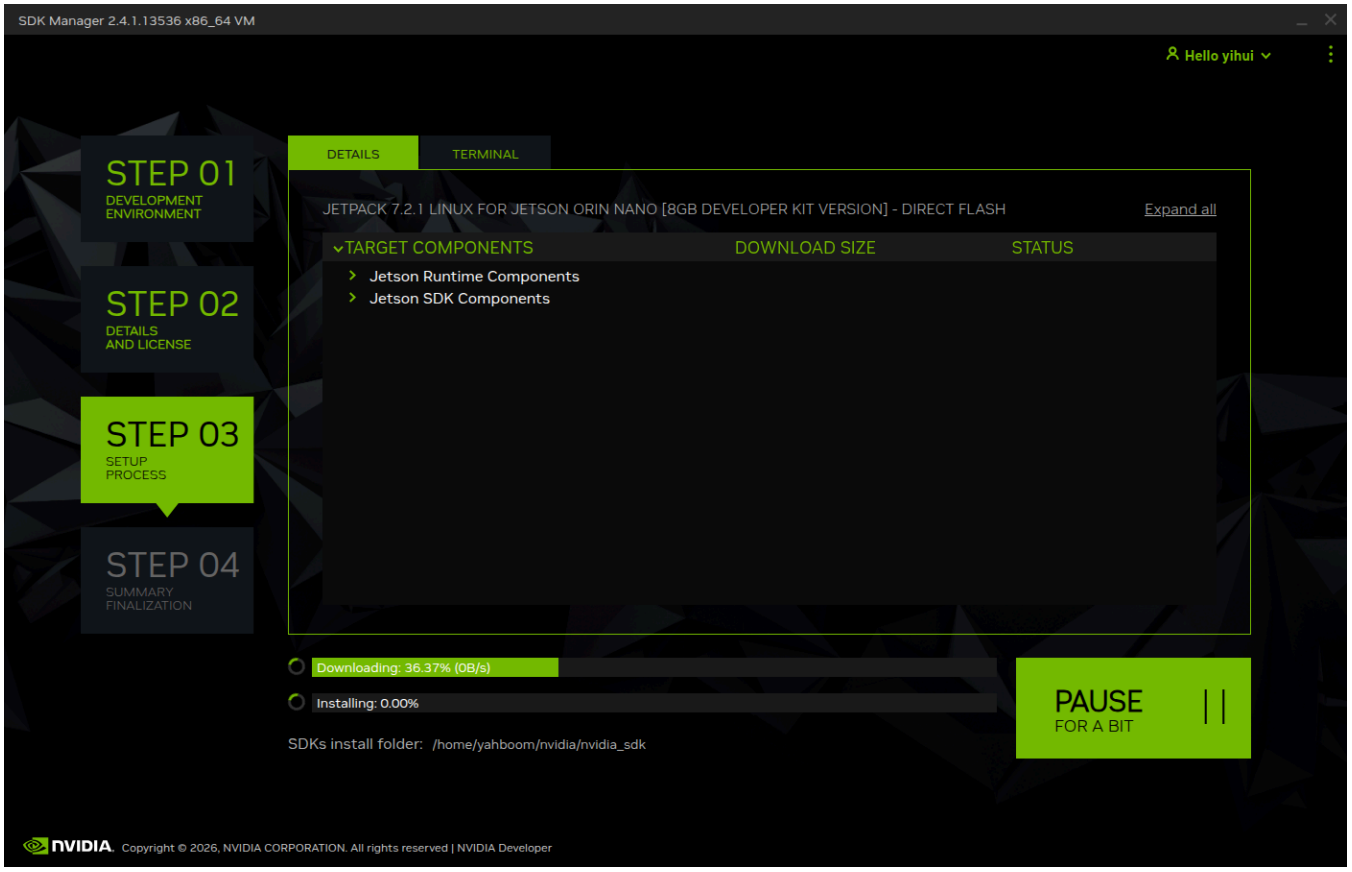
Here enter the user name and password you set yourself when the system was first booted: the whole process takes a while, so please be patient

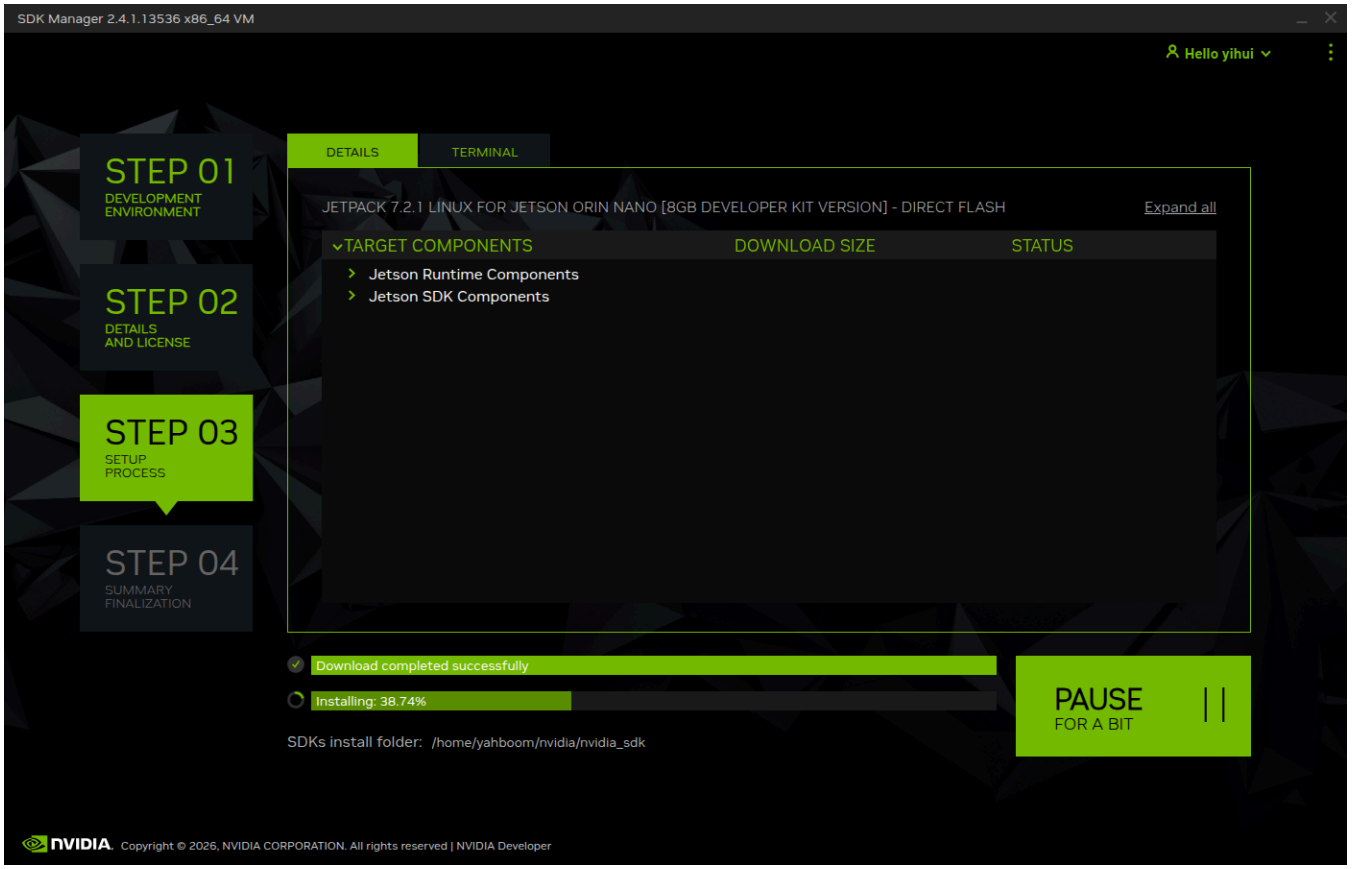


Wait for the system to download and install: network failures may appear in the virtual machine during the whole process; you can ignore them, because in practice they do not affect the flashing of the component environment

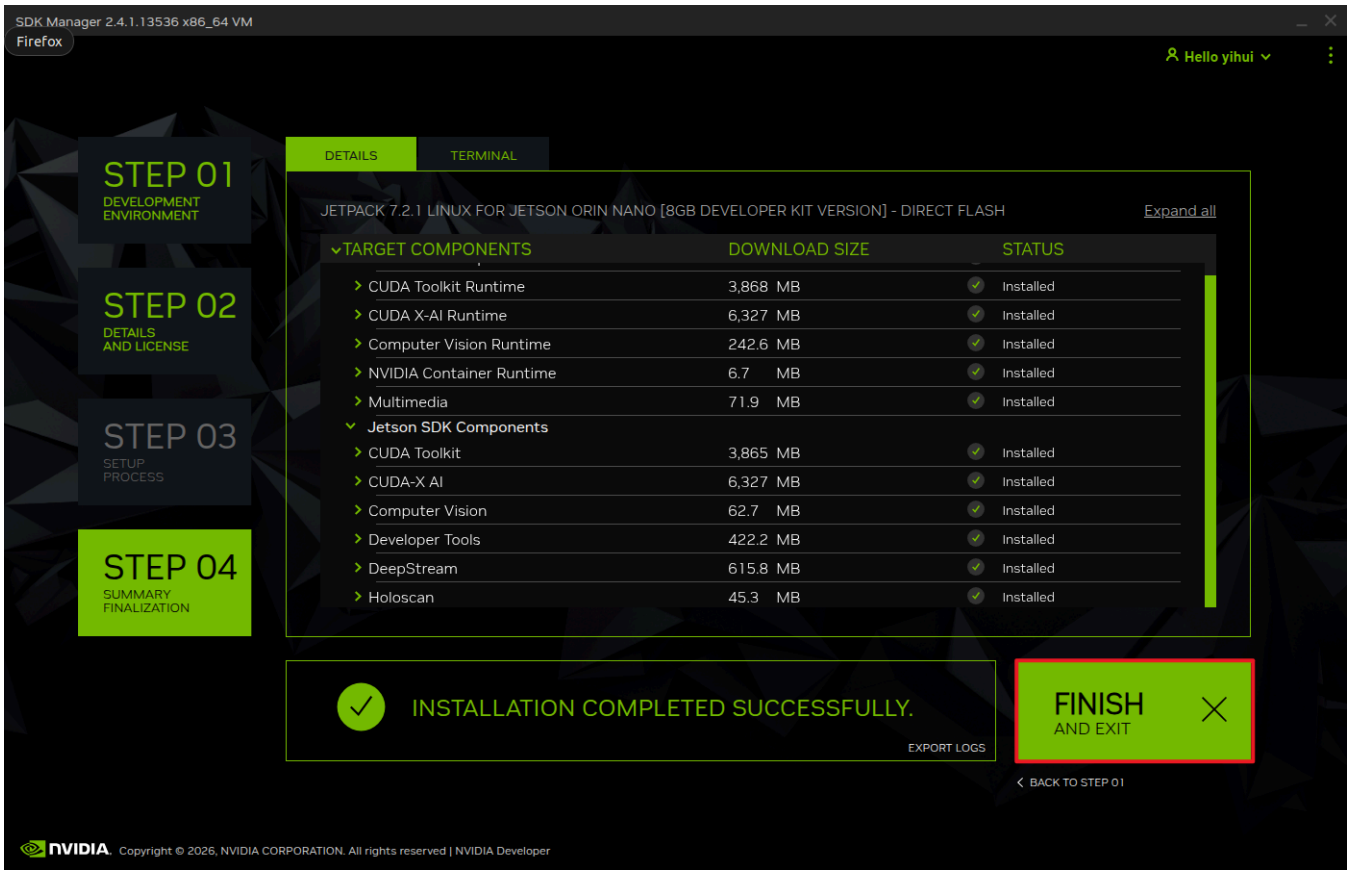


The whole installation process takes a long time and requires patience; if the installation fails, you can reboot the mainboard system and restart the SDK Manager installation:





3.5. STEP4

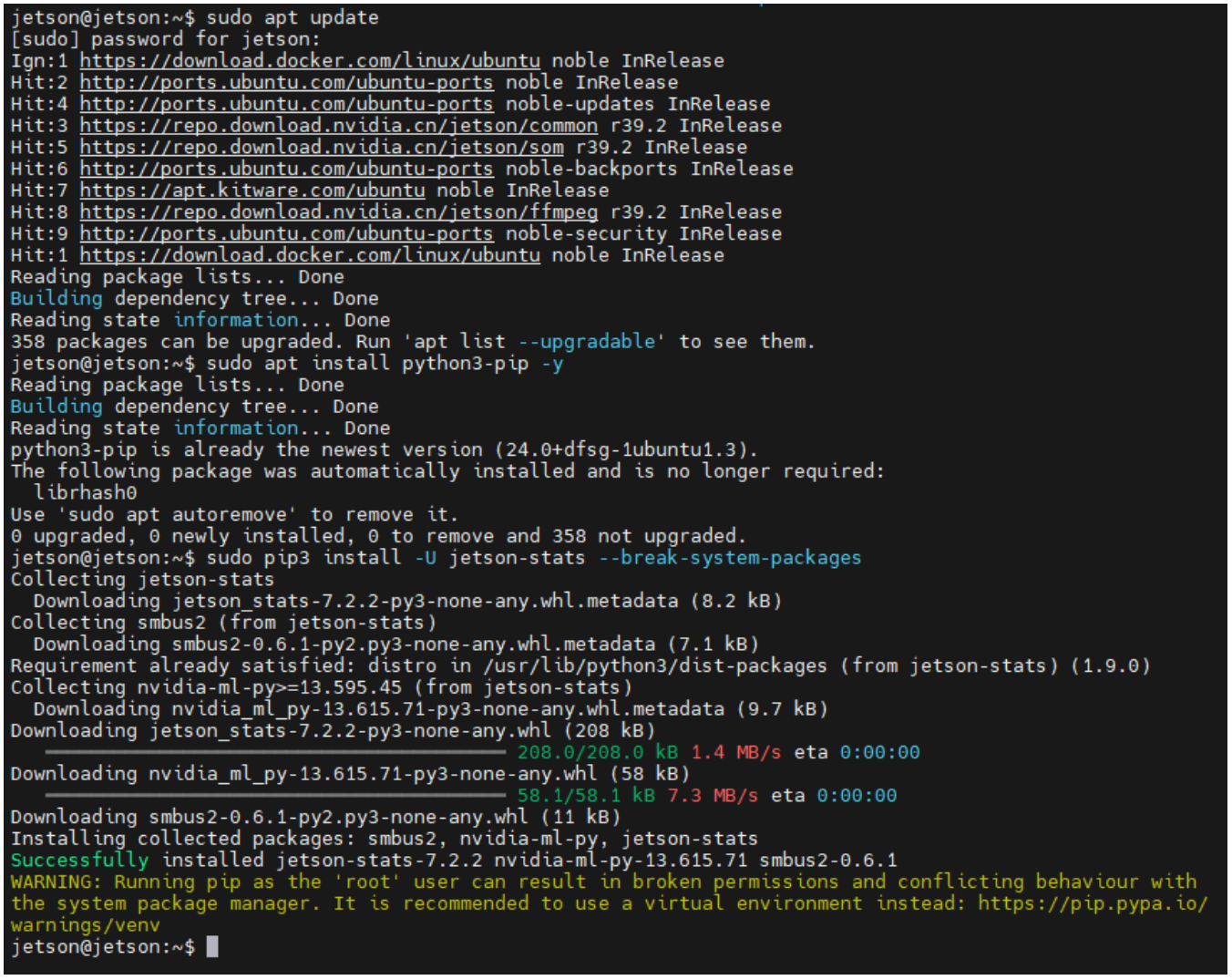


4. Environment verification

> **Jtop tool**

Install Jtop with the following command:

```
sudo apt update
sudo apt install python3-pip -y
sudo pip3 install -U jetson-stats --break-system-packages
sudo jtop --install-service
```



Note: The system must be rebooted after the installation!

> **Maximum performance mode**

Enable the MAX power mode

```
sudo nvpmodel -m 0 # Jetson Orin NX
```

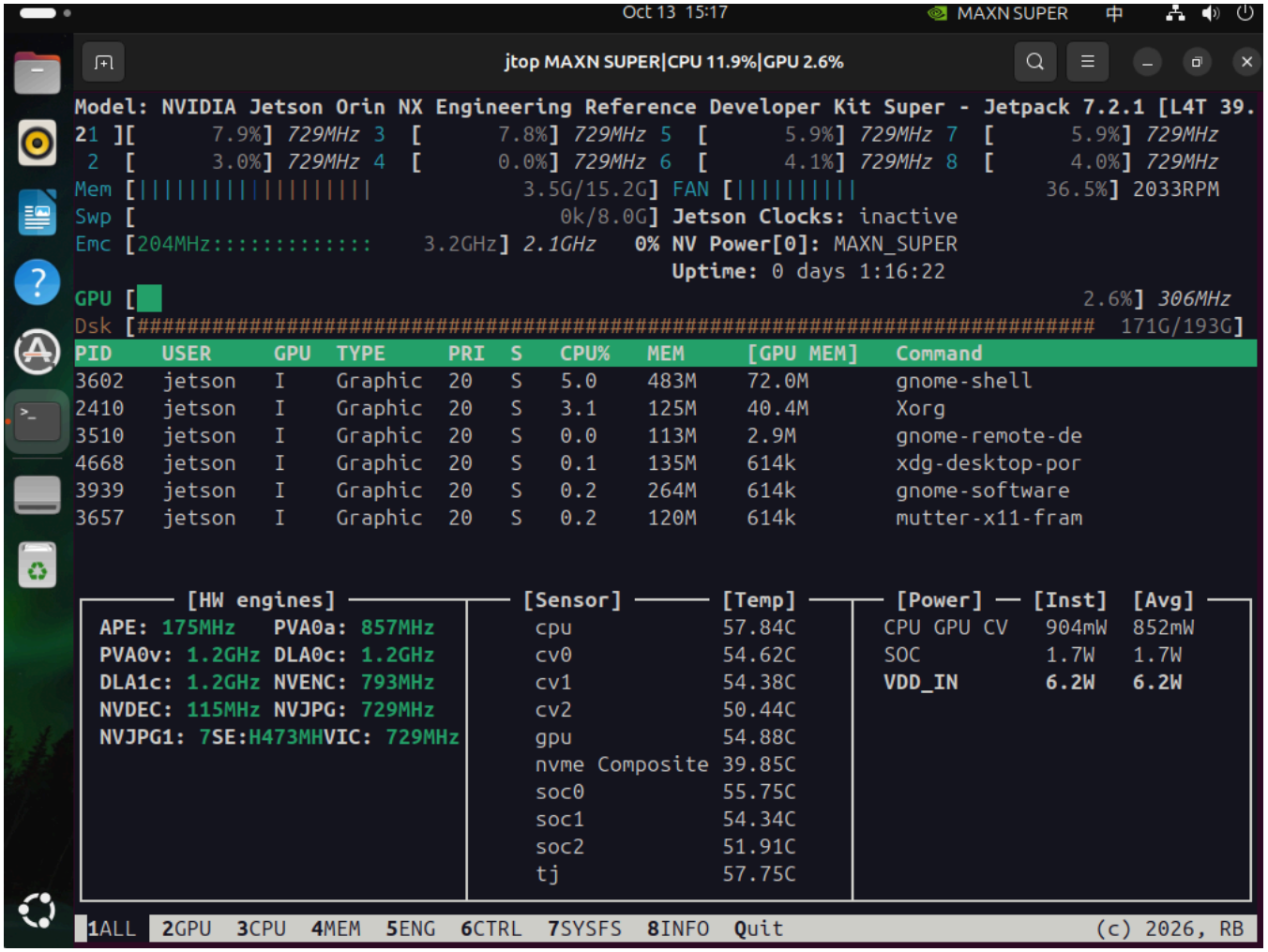


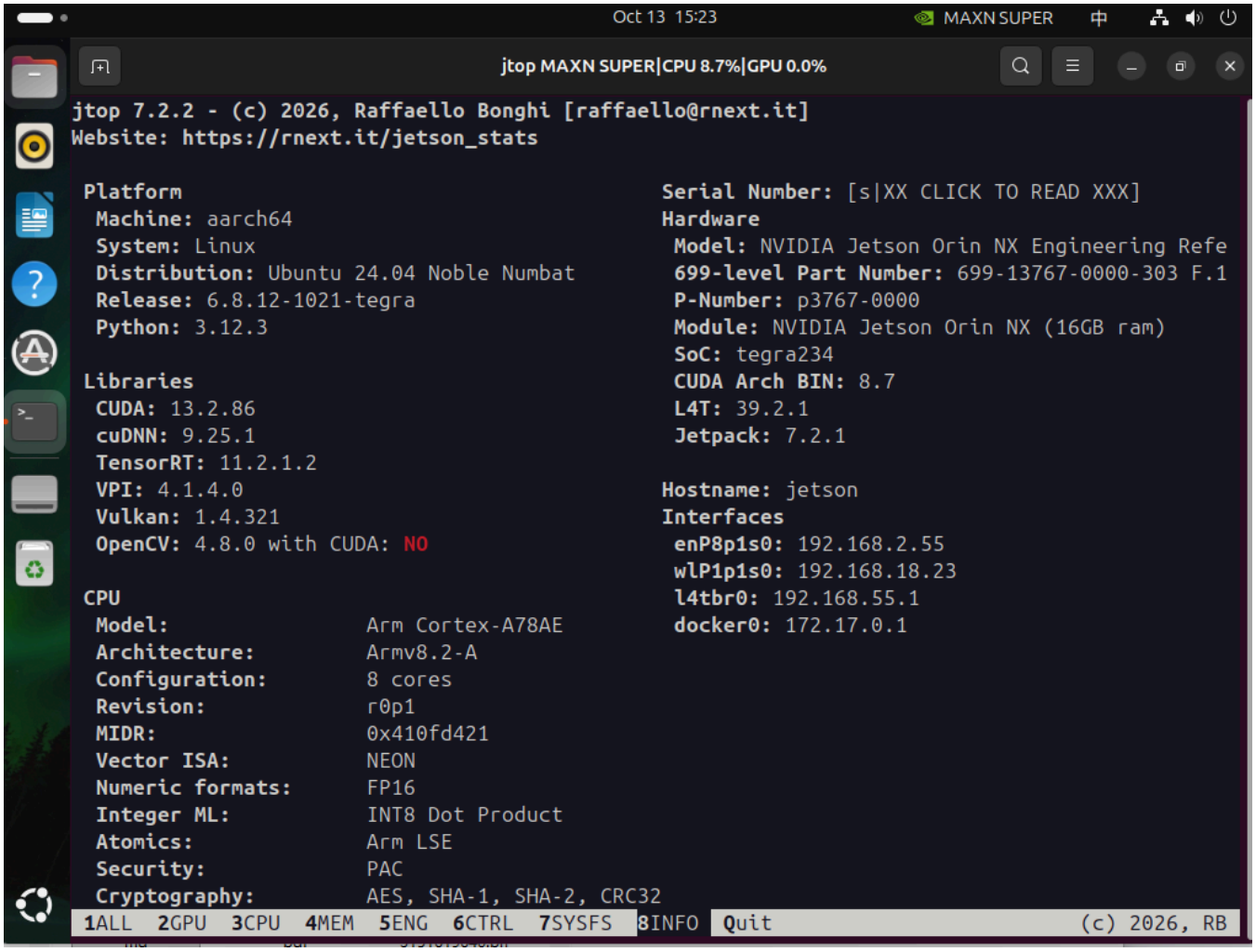
Enable the Jetson clocks: the CPU and GPU cores run at the maximum frequency

```
sudo jetson_clocks
```

Use the Jtop tool to view the system information:

```
jtop
```


Oct 13 15:17
MAXN SUPER
jtop MAXN SUPER|CPU 11.9%|GPU 2.6%
Model: NVIDIA Jetson Orin NX Engineering Reference Developer Kit Super - Jetpack 7.2.1 [L4T 39.
21][7.9%] 729MHz 3 [7.8%] 729MHz 5 [5.9%] 729MHz 7 [5.9%] 729MHz
2 [3.0%] 729MHz 4 [0.0%] 729MHz 6 [4.1%] 729MHz 8 [4.0%] 729MHz
Mem [3.5G/15.2G] FAN [36.5%] 2033RPM
Swp [0k/8.0G] Jetson Clocks: inactive
Emc [204MHz: 3.2GHz] 2.1GHz 0% NV Power[0]: MAXN_SUPER
Uptime: 0 days 1:16:22
GPU [2.6%] 306MHz
Dsk [171G/193G]
PID USER GPU TYPE PRI S CPU% MEM [GPU MEM] Command
3602 jetson I Graphic 20 S 5.0 483M 72.0M gnome-shell
2410 jetson I Graphic 20 S 3.1 125M 40.4M Xorg
3510 jetson I Graphic 20 S 0.0 113M 2.9M gnome-remote-de
4668 jetson I Graphic 20 S 0.1 135M 614k xdg-desktop-por
3939 jetson I Graphic 20 S 0.2 264M 614k gnome-software
3657 jetson I Graphic 20 S 0.2 120M 614k mutter-x11-fram
[HW engines]
APE: 175MHz PVA0a: 857MHz
PVA0v: 1.2GHz DLA0c: 1.2GHz
DLA1c: 1.2GHz NVENC: 793MHz
NVDEC: 115MHz NVJPG: 729MHz
NVJPG1: 7SE:H473MHVIC: 729MHz
[Sensor] [Temp]
cpu 57.84C
cv0 54.62C
cv1 54.38C
cv2 50.44C
gpu 54.88C
nvme Composite 39.85C
soc0 55.75C
soc1 54.34C
soc2 51.91C
tj 57.75C
[Power] [Inst] [Avg]
CPU GPU CV 904mW 852mW
SOC 1.7W 1.7W
VDD_IN 6.2W 6.2W
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit (c) 2026, RB


Oct 13 15:23
MAXN SUPER
jtop MAXN SUPER|CPU 8.7%|GPU 0.0%
jtop 7.2.2 - (c) 2026, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats
Platform
Machine: aarch64
System: Linux
Distribution: Ubuntu 24.04 Noble Numbat
Release: 6.8.12-1021-tegra
Python: 3.12.3
Libraries
CUDA: 13.2.86
cuDNN: 9.25.1
TensorRT: 11.2.1.2
VPI: 4.1.4.0
Vulkan: 1.4.321
OpenCV: 4.8.0 with CUDA: NO
CPU
Model: Arm Cortex-A78AE
Architecture: Armv8.2-A
Configuration: 8 cores
Revision: r0p1
MIDR: 0x410fd421
Vector ISA: NEON
Numeric formats: FP16
Integer ML: INT8 Dot Product
Atomics: Arm LSE
Security: PAC
Cryptography: AES, SHA-1, SHA-2, CRC32
Serial Number: [s|XX CLICK TO READ XXX]
Hardware
Model: NVIDIA Jetson Orin NX Engineering Refe
699-level Part Number: 699-13767-0000-303 F.1
P-Number: p3767-0000
Module: NVIDIA Jetson Orin NX (16GB ram)
SoC: tegra234
CUDA Arch BIN: 8.7
L4T: 39.2.1
Jetpack: 7.2.1
Hostname: jetson
Interfaces
enP8p1s0: 192.168.2.55
wlP1p1s0: 192.168.18.23
l4tbr0: 192.168.55.1
docker0: 172.17.0.1
1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7SYSFS 8INFO Quit
(c) 2026, RB